VIC-ML-KEM-X Post-Quantum Cryptography Core

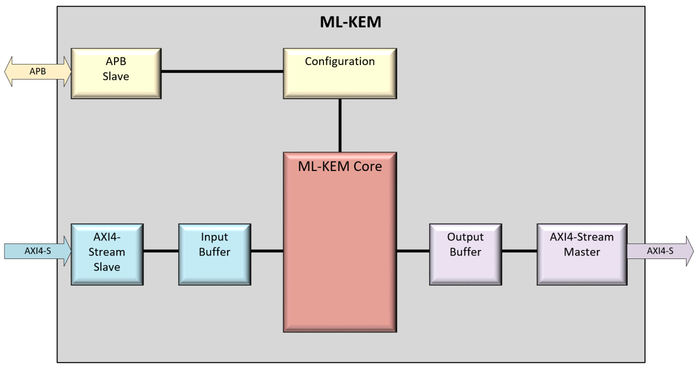
Ultra-Lean, ACVP-Verified, SCA-Protected IP Core for FIPS 203 Compliance

Overview

The **VIC-ML-KEM-X** is a high-performance, production-ready hardware IP core implementing the finalized NIST FIPS 203 ML-KEM standard. Built entirely as a pure, monolithic hardware state machine (no hidden RISC-V CPU or firmware dependency), it delivers up to 50x faster execution than soft-processor implementations while occupying an ultra-compact **12.34k LUT** envelope.

Designed for security-critical IoT and automotive platforms, the core combines zero-overhead monolithic execution with **multi-layer Side-Channel Analysis (SCA) resistance** across its symmetric, arithmetic, and decapsulation pipelines.

Block Diagram



Key Features & Benefits

- **100% ACVP Verified & FIPS 203 Compliant:** Fully passes all official NIST ACVP projection vector suites for KeyGen, Encapsulation, and Decapsulation across ML-KEM-512, 768, and 1024 profiles.
- **Built-in Algorithmic SCA Protection:** Multi-layered physical security guarding against DPA, CPA, and Chosen-Ciphertext attacks without incurring traditional "SCA tax" LUT explosions.
- **Pure Monolithic RTL Architecture**: 100% self-contained hardware engine—requires no embedded microcontroller, external software stack, or toolchain integration.
- **Ultra-Lean Silicon Footprint:** Fits inside **12,341 LUTs** and **6 DSPs** (512 profile), preserving scarce logic and math tiles on cost-sensitive FPGAs and ASICs.
- **High-Frequency Operation:** Reaches a **6.25 ns clock period (160 MHz)** on entry-level FPGAs, enabling seamless integration into high-speed bus domains.
- **Native Montgomery Domain Processing:** Retains polynomial coefficients in the Montgomery domain throughout NTT/iNTT pipelines, eliminating redundant reduction steps.
- **Zero Third-Party IP Dependencies:** Built 100% from the ground up by Vicip for clean provenance, straightforward export compliance, and unambiguous legal indemnification.

Multi-Layer Side-Channel Protection Architecture

The core incorporates localized, low-overhead algorithmic hiding and masking countermeasures targeting power and electromagnetic (EM) side-channel leakage:

1. **Symmetric Engine Time Desynchronization & Resilient PRNG:** Integrated pseudo-random clock cycle stalling driven by an internal PRNG/LFSR within the Keccak (SHAKE) engine to defeat trace-alignment attacks. The PRNG utilizes continuous additive state-mixing (XOR accumulation), ensuring that software seed updates continuously mutate internal entropy to prevent replay, software-forced static traces, or deterministic trace alignment.
2. **Point-Wise Multiplication (PWM) Execution Shuffling:** Dynamic randomization of polynomial coefficient multiplication sequences in the frequency domain, desynchronizing power traces to break

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Correlation Power Analysis (CPA) on secret-key math.

3. **SCA-Protected Decapsulation Comparator (FO Transform):** The combined streaming Compress-Encode-Compare pipeline features a two-layer defense against Fujisaki-Okamoto leakage:
    - **Input Masking:** Incoming and expected ciphertext bytes are XOR-masked with fresh dynamic randomness prior to comparison.
    - **2-Share Boolean Accumulator:** Tracks mismatch state using dual XOR-derived Boolean shares updated on *every* clock cycle, ensuring constant dynamic power consumption regardless of match status.

Performance & Utilization Summary

Metrics represent an optimized implementation on AMD/Xilinx 7-Series targeting the ML-KEM-512 profile (160 MHz / 6.25 ns).

Resource Type	Utilized
Slice LUTs	12,341
Slice Registers	7,678
Dedicated DSP Blocks	6
Internal RAM / Block RAM	4.8 kB 4.5 BRAM
Clock Period / Frequency	6,25 ns 160 MHz

Cryptographic Operation	Execution Cycles/Time
Key Generation	10,560 cycles/66 µs
Encapsulation	14,240 cycles/89 µs
Decapsulation	19680 cycles/123 µs

Technical Specifications

Control & Data Plane Interfaces

- **Control Plane (APB Slave):** AMBA APB interface for command submission, status monitoring, and runtime configuration.
- **Data Plane (AXI4-Stream):** High-throughput streaming interface for seeds, public/private keys, and ciphertext offloading.
- **Configurability:** Available as pre-synthesis standalone cores (512, 768, or 1024) or APB-configurable multi-mode engines (Dual 768/512 or Tri-Mode 1024/768/512) for software runtime switching.

Primary Applications

- **Post-Quantum Secure Boot:** Validating epistemic root key exchanges during low-power MCU boot sequences.
- **Automotive MCU Security:** Low-latency protection for critical V2X communications and ECU networks.
- **Industrial IoT Gateways:** Compact edge hardware allowing constrained devices to establish post-quantum TLS/DTLS sessions.
- **Space-Grade Protocol Integration:** Compact footprint fits radiation-tolerant FPGAs to combat "Harvest Now, Decrypt Later" threats on satellite RF downlinks.

Deliverables & Getting Started

Vicip delivers full synthesizable Verilog/SystemVerilog RTL, automated self-checking simulation suites mapping to official NIST ACVP vectors, gate-level simulation (GLS) scripts, and integration documentation.

Contact Vicip today for evaluation licensing, ASIC node synthesis results, or custom configurations.

Disclaimer: *ML-KEM is a NIST-standardized algorithm. This IP core is a hardware implementation of that algorithm. Performance and utilization figures are representative and will vary based on target technology, synthesis tools, and configuration settings.*